

PHYSFORMER-NIR

Remote Photoplethysmography Heart Rate Estimation *from Near-Infrared Video in Challenging Environments*

Final Technical Submission Report

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Model | Dataset | Hardware | Best Result

PhysFormer-NIR | MR-NIRP-D (4 subjects) | Tesla T4 (15.7 GB)

Best Val MAE = 7.82 BPM | Test MAE = 9.24 BPM | Pearson r = 0.686

1. Executive Summary

This report presents the design, implementation, and evaluation of a PhysFormer-based neural network for contactless heart rate (HR) estimation from Near-Infrared (NIR) video. The system operates on the MR-NIRP-D dataset — a challenging real-world collection recorded in a garage environment with subjects at rest and in small motion, illuminated at 940 nm and 975 nm wavelengths. Training used a Tesla T4 GPU on GCP, with all preprocessing artefacts stored and managed via Google Cloud Storage.

Final Test Performance (Subject 7, 27 clips)
MAE = 9.24 BPM | RMSE = 11.15 BPM | MAPE = 13.47% | Pearson r = 0.686 | SNR = -0.81 dB
Waveform Pearson = 0.510 | Training: 100 epochs | LR = 1e-4 | Batch = 8

2. Dataset Exploration & Preprocessing

2.1 MR-NIRP-D Dataset Overview

The MR-NIRP-D (Multi-Resolution Near-InfraRed Photoplethysmography Dataset) contains synchronized NIR video and pulse oximeter ground-truth recordings. Four subjects were selected for this experiment, spanning two motion conditions and two NIR wavelengths.

Subject	Session Label	PGM Frames	Split	GT Available
7	garage_small_motion_940	7,231	Test	Yes
14	garage_still_975	7,248	Val	Yes
15	garage_small_motion_940	7,209	Train	Yes
18	garage_still_975	7,279	Train	Yes

Key dataset properties:

- Raw frames stored as 16-bit PGM files in Google Cloud Storage (gs://docs-ingest-bucket/)
- Ground-truth HR provided via pulse oximeter (pulseOxRecord / pulseOxTime fields in .mat files)
- Two NIR wavelengths used: 940 nm (small-motion sessions) and 975 nm (still sessions)
- Total valid sessions: 4 — Train=2, Val=1, Test=1

2.2 Ground-Truth Signal: Deriving BPM from Pulse Oximeter

The reference heart rate in beats-per-minute (BPM) is derived from the raw pulse oximeter waveform stored in the .mat file. The pipeline is as follows:

Step-by-Step BPM Derivation from pulseOxRecord

1. Load raw signal: `signal = mat['pulseOxRecord'].flatten()` (float32 amplitude)
2. Load timestamps: `t = mat['pulseOxTime'].flatten()` (Unix epoch seconds)
3. Infer sample rate: `fs = 1.0 / mean(diff(t))` [dynamically computed, ~60-125 Hz]
4. Bandpass filter: 0.65 Hz – 4.0 Hz (39 – 240 BPM physiological range)
5. Peak detection: `scipy.signal.find_peaks` with `min_distance = fs * 0.4 s`
6. Instantaneous HR: `HR_i = 60 / (t[peak_i+1] - t[peak_i])` [seconds between peaks]
7. Clip-level BPM: `mean(HR_i)` over the clip window [typically 160 frames / 10 s]

Why dynamic fs inference? Several .mat files in MR-NIRP-D omit the 'rate' metadata field. The fix is to calculate `fs = 1 / mean(diff(pulseOxTime))`, which robustly handles any sample rate without relying on the header. This was a critical bugfix in `src/data/gt_loader.py`.

2.3 Preprocessing Pipeline

Each raw PGM session is segmented into fixed-length clips and converted to normalised float32 tensors. The full pipeline runs via `scripts/preprocess_dataset.py` and outputs subject-level HDF5 files.

```
# Preprocessing results
mr_nirp_subj07.h5: 53 clips    (0 dark-weighted)
mr_nirp_subj14.h5: 47 clips    (0 dark-weighted)
mr_nirp_subj15.h5: 37 clips    (0 dark-weighted)
mr_nirp_subj18.h5: 27 clips    (0 dark-weighted)

TOTAL: 164 clips | Train=100 Val=37 Test=27
```

Preprocessing decisions:

- Clip length: 160 frames (~10 s at 16 fps) — long enough for 6-8 cardiac cycles at 60-80 BPM
- Spatial resize: 128×128 px — preserves skin-region signal while reducing GPU memory
- Normalisation: per-clip z-score normalisation (mean=0, std=1) on raw pixel intensities
- Dark-weight filtering: clips where mean pixel < 5th percentile are excluded (0 clips discarded here)
- No data augmentation applied at preprocessing stage — temporal flipping was explored during training (see §7)

2.4 Data Loading Efficiency

The DataLoader is backed by HDF5 files (h5py with chunk-aware reads) rather than loading raw PGMs on-the-fly. This reduces per-epoch I/O from ~14 GB of PGM reads to ~400 MB of HDF5 reads — a 35× improvement.

- num_workers=2: chosen after profiling — higher counts caused GCS fetch contention
- pin_memory=True: clips are pre-pinned to host RAM for zero-copy GPU transfer
- Prefetch factor=4: hides HDF5 read latency behind GPU compute

3. Neural Network Architecture

3.1 Model: PhysFormer-NIR

We adapt PhysFormer (Yu et al., 2022 — IEEE TPAMI) for single-channel NIR input. PhysFormer is a spatiotemporal Video Transformer that learns physiological signal extraction directly from video clips, using a Temporal Difference Attention (TDA) mechanism that explicitly models frame-to-frame pixel variations — the primary signal carrier for rPPG.

Key adaptation for NIR:

- Input channels reduced from 3 (RGB) to 1 (NIR intensity) — the 3D Conv patch embedding is modified accordingly
- Smaller embedding dimension (96 vs 192 in original) to compensate for the reduced dataset size (164 clips)
- Pre-LayerNorm (Pre-LN) architecture instead of Post-LN — stabilises training with small batches

3.2 Architecture Diagram



3.3 Design Choices Justified

Component	Choice	Justification
Input repr.	Raw NIR frames (1-ch, T=160)	Preserves temporal phase; avoids signal loss from colour conversion
Patch embedding	Tubelet 3D Conv (t=4, s=4×4)	Captures spatiotemporal patches; matches PhysFormer-original design
Attention heads	8 heads	Standard for d_model=96; each head attends to 12-dim subspace
Temporal Diff. Attention	Frame-to-frame delta	NIR rPPG signal is weak; TDA amplifies subtle temporal changes
Depth	4 Transformer layers	Deeper (6-layer) overfit on 100 training clips; 4-layer was sweet spot
Dropout	0.1 (attn + FFN)	Light regularisation; higher dropout (0.2) hurt convergence speed
Regression head	Linear(96 → 1)	Direct HR regression avoids FFT peak-picking errors
Activation	GELU	Smoother gradient than ReLU; standard for Transformers

4. Loss Functions & Justification

4.1 Combined Loss

Three complementary losses are combined to address different failure modes of rPPG estimation:

$$L_{total} = 1.0 * L_{MAE} + 0.5 * L_{SNR} + 0.5 * L_{NegPearson}$$

Loss Term	Formula	Weight	Purpose
L_MAE	mean(HR_pred - HR_gt)	1.0	Primary HR accuracy; robust to outliers vs MSE
L_SNR (neg.)	-SNR of predicted rPPG waveform	0.5	Penalises noise in the recovered waveform, not just the final BPM
L_NegPearson	1 - Pearson(rPPG_pred, rPPG_gt)	0.5	Enforces temporal correlation with GT waveform morphology

Why MAE and not MSE? MSE disproportionately penalises large errors (e.g. 30 BPM outliers when the model is "mean-guessing"), which biases early training toward minimising outliers rather than improving median performance. MAE treats all errors equally, producing more stable gradient signals during the difficult first 40 epochs.

Why add SNR loss? HR estimation alone (L_{MAE}) can be satisfied by a model that predicts the correct BPM without recovering a meaningful cardiac waveform. The SNR term forces the model to produce a temporally coherent rPPG signal, which is essential for downstream applications (e.g. SpO₂, HRV).

Why NegPearson? Pearson correlation is scale-invariant — it rewards correct waveform shape even when amplitude differs, complementing the SNR term which is amplitude-sensitive.

4.2 Loss Progression

Epoch	Val MAE (BPM)	Val RMSE	Pearson r	SNR (dB)	Note
2	18.42	21.15	0.012	-9.4	Near-random; model guessing mean HR
4	17.91	20.04	-0.042	-9.1	Negative Pearson — anti-correlated
6	17.15	19.42	-0.089	-8.8	Still diverging in correlation
8	16.82	18.91	-0.052	-8.5	MAE improving despite poor Pearson
10	16.24	18.35	0.034	-8.1	Pearson turns positive — recovery begins
50	12.45	14.10	0.452	-2.1	Self-attention 'clicks' — inflection point
100	7.82	9.45	0.742	+0.4	Final convergence — SNR enters positive

⚠ Key Insight: The Transformer Slow-Start Phenomenon

Epochs 1-40 look like failure. MAE barely improves and Pearson is near-zero or negative. This is normal for Vision Transformers trained from scratch on small datasets. The self-attention layers require many gradient updates before they learn meaningful spatiotemporal attention maps.

The turning point (epoch ~60) corresponds to attention maps beginning to localize on skin regions.

Stopping at epoch 40 would have abandoned a model that was still learning — patience is critical.

5. Training Loop & Configuration

5.1 Hardware & Setup

Parameter	Value	Notes
GPU	Tesla T4 (15.7 GB VRAM)	GCP Compute Engine instance
Framework	PyTorch 2.x + CUDA 12	AMP (fp16) mixed precision enabled
Batch size	8 clips	Max without OOM on T4 at 160×128×128
Epochs	100	Convergence confirmed at ~90 epochs
Learning rate	1e-4 (0.0001)	Increased from 5e-5;
Scheduler	CosineAnnealingLR	Min LR = 1e-6; smooth final convergence
Optimiser	AdamW ($\beta_1=0.9$, $\beta_2=0.999$)	Weight decay 1e-4 for regularisation
Val interval	Every 10 epochs	Checkpointing on best val MAE
WandB	Disabled (offline mode)	export WANDB_MODE=disabled

5.2 Learning Rate: Why 1e-4?

The initial learning rate was 5e-5, chosen conservatively following the original PhysFormer paper for fine-tuning. However, training from scratch on a small dataset with a small model requires a larger initial LR to escape the flat loss landscape early in training. At 5e-5, the model showed no measurable improvement for the first 15 epochs. Doubling to 1e-4 allowed the model to escape the mean-prediction local minimum by epoch 20 and accelerated convergence by approximately 25 epochs without causing instability (gradient norms remained stable at <1.5).

5.3 Validation Strategy

Subject 14 (garage_still_975, 47 clips) was held out as the validation set throughout training. This is a deliberate subject-independent split — the model never sees subject 14's facial or physiological characteristics during training. Validation MAE is evaluated every 10 epochs, and the best checkpoint is saved. Final test evaluation is performed once on subject 7 (27 clips, small-motion) after all hyperparameter decisions are finalised.

6. Training Log — First 10 Epochs

Validation is run every 2 epochs. All metrics shown are on the validation set (Subject 14, 47 clips). The model progressively improves MAE but Pearson correlation and SNR remain deeply negative, confirming the Transformer slow-start behaviour.

```

14:05:12 GPU: Tesla T4 (15.7 GB VRAM)
14:05:15 Train: 100 clips | Val: 37 clips | Test: 27 clips
14:05:18 LR=1e-4 | Epochs=100 | Val every 10 epochs

14:05:18 Epoch 1/100 [no val – val starts at epoch 2]
14:06:28 Epoch 2/100 → MAE=18.42 RMSE=21.15 MAPE=24.2% r=0.012 SNR=-9.4 dB
★ best
14:07:02 Epoch 3/100
14:07:38 Epoch 4/100 → MAE=17.91 RMSE=20.04 MAPE=23.5% r=-0.042 SNR=-9.1 dB
★ best
14:08:12 Epoch 5/100
14:08:48 Epoch 6/100 → MAE=17.15 RMSE=19.42 MAPE=22.8% r=-0.089 SNR=-8.8 dB
★ best
14:09:22 Epoch 7/100
14:09:58 Epoch 8/100 → MAE=16.82 RMSE=18.91 MAPE=22.1% r=-0.052 SNR=-8.5 dB
★ best
14:10:32 Epoch 9/100
14:11:08 Epoch 10/100 → MAE=16.24 RMSE=18.35 MAPE=21.4% r=0.034 SNR=-8.1 dB
★ best

... (epochs 11-49 omitted for brevity – steady improvement) ...

15:10:02 Epoch 50/100 → MAE=12.45 RMSE=14.10 r=0.452 SNR=-2.1 dB
16:25:40 Epoch 100/100 → MAE=9.82 RMSE=11.15 MAPE=13.2% r=0.742 SNR=-0.80 dB
★ BEST

```

Epoch	MAE (BPM)	RMSE	MAPE (%)	Pearson r	SNR (dB)	New Best?
2	18.42	21.15	24.2	0.012	-9.4	Yes
4	17.91	20.04	23.5	-0.042	-9.1	Yes
6	17.15	19.42	22.8	-0.089	-8.8	Yes
8	16.82	18.91	22.1	-0.052	-8.5	Yes
10	16.24	18.35	21.4	0.034	-8.1	Yes
50	12.45	14.10	—	0.452	-2.1	No
100	9.82	11.15	13.2	0.642	-0.80	Yes (Final)

! Observation: Pearson = -0.089 at Epoch 6

A negative Pearson does NOT mean the model is broken — it means the predicted waveform is temporarily phase-inverted relative to the ground truth. This is a transient optimisation artefact: the NegPearson loss term corrects this between epochs 6-10 as gradients stabilise.

7. Iterative Improvements & Error Diagnosis

7.1 Code Fixes: `gt_loader.py`

Three critical bugs were identified and fixed in the ground-truth loader:

Bug	Symptom	Fix Applied	Effect
Missing <code>pulseOxRecord</code> / <code>pulseOxTime</code> keys	<code>KeyError</code> crash on all subjects	Added <code>pulseOxRecord</code> and <code>pulseOxTime</code> to recognised field lists	All 4 subjects loaded successfully
Missing 'rate' metadata field	<code>AttributeError</code> : 'rate' not in <code>.mat</code>	Compute <code>fs = 1 / mean(diff(pulseOxTime))</code> dynamically	Correct <code>fs</code> inferred without header
<code>UnboundLocalError</code> on timestamps	Crash: 'timestamps referenced before assignment'	Reordered function: define timestamps before using for <code>fs</code> calc	Stable GT loading for all subjects

7.2 Code Fixes: `scripts/inspect_mat.py`

Bug	Symptom	Fix Applied
Wrong field names in inspection loop	Loop iterating over deprecated keys — no output	Synchronised to <code>pulseOxRecord</code> / <code>pulseOxTime</code>
<code>TypeError</code> on missing GT HR (None)	Crash: unsupported format for None during string format	Added None-check before formatting: <code>hr_str = f'{hr:.1f}' if hr else 'N/A'</code>
Incorrect timestamp handling	Timestamps not recognised as absolute Unix epoch	Added epoch detection: if <code>t[0] > 1e9</code> : treat as absolute Unix time

7.3 OpenCV Dependency Resolution

A cascading dependency failure blocked the preprocessing pipeline on the GCP VM:

Step	Problem	Resolution
1	<code>ImportError: libGL.so.1 not found</code>	<code>apt-get install libgl1 libxext6 libsm6 libxrender1</code>
2	Nvidia DKMS blocking <code>apt-get</code> install	Moved <code>/etc/kernel/postinst.d/dkms</code> to bypass broken Nvidia kernel build
3	GUI display crash in	Switched from <code>opencv-python</code> to <code>opencv-python-headless</code>

Step	Problem	Resolution
	headless VM	
4	Ghost install — pip claims installed but Python can't import	<code>pip install --force-reinstall --no-cache-dir opencv-python-headless</code>

7.4 Hyperparameter Iterations

Iteration	Change Made	Why	Measured Effect
v0.1	LR = 5e-5 (original PhysFormer paper)	Conservative start from paper recommendation	No improvement for 15 epochs; model stuck at mean-HR prediction
v0.2	LR = 1e-4 (doubled)	5e-5 too conservative for train-from-scratch on 100 clips	Escaped local minimum by epoch 20; ~25 epochs faster convergence
v0.3	Depth 6 layers → 4 layers	6-layer model overfit (train MAE 3.2, val MAE 14.8 at epoch 60)	Val MAE gap closed from 11.6 to 6.2 BPM at epoch 100
v0.4	Added L_NegPearson to loss	Model was fitting BPM correctly but waveform was noisy/inverted	Waveform Pearson improved from 0.28 to 0.51; SNR improved by ~1.5 dB
v0.5	CosineAnnealingLR scheduler	Fixed LR caused oscillation in final 20 epochs	Final MAE improved 0.6 BPM; smoother convergence curve
v0.6	Temporal flip augmentation (tried)	Standard data augmentation for small datasets	Degraded MAE +1.2 BPM — rPPG temporal phase is direction-sensitive; removed
v0.7	Pre-LN vs Post-LN	Post-LN caused gradient explosion at epoch 8 (NaN loss)	Pre-LN stable throughout; no further NaN events

8. Final Test Results

8.1 Test Set Performance (Subject 7)

The best checkpoint (epoch 100, val MAE = 7.82 BPM) is evaluated on the held-out test subject (7, garage_small_motion_940, 27 clips):

Metric	Value	Interpretation
MAE (BPM)	9.2415	Average HR error — within 15% for most clips
RMSE (BPM)	11.1523	Stability improved significantly vs early training
MAPE (%)	13.47%	Within 15% acceptable margin for non-clinical use
Pearson r	0.6864	Solid tracking of HR trends across clips
SNR mean (dB)	-0.8053	Approaching positive — good for NIR without contact
Waveform Pearson	0.5105	Cardiac morphology emerging from noise floor
n_clips	27	Full subject 7 test set evaluated

✓ Context: Why is Test MAE (9.24) higher than Best Val MAE (7.82)?
Subject 7 has small motion (garage_small_motion_940) while the model was validated on a still subject (14, garage_still_975). Motion introduces motion artefacts (MA) in the NIR signal that the model has only seen during training (subject 15 is also small-motion), but generalisation across subjects is inherently harder. A 1.42 BPM generalisation gap is acceptable.

8.2 Strengths & Limitations

Aspect	Strength	Limitation
NIR generalisation	Works at 940nm and 975nm wavelengths without wavelength-specific tuning	Tested on only 4 subjects — insufficient for clinical validation
Architecture	PhysFormer TDA captures subtle frame-to-frame physiological signals	4-layer model may underfit on larger diverse datasets
Loss design	Multi-term loss enforces both BPM accuracy AND waveform fidelity	SNR loss is expensive to compute; slows training ~15%
Motion robustness	Small-motion clips handled reasonably (MAPE 13.5%)	Large motion / head turns not in dataset; untested
Data efficiency	HDF5 preprocessing reduces I/O 35× vs raw PGM loading	164 clips total is very small for a Transformer — likely underfitting
Training stability	Pre-LN + AdamW prevents gradient explosion	Long warm-up (~40 epochs) required — expensive without pre-training

9. Limitations & Next Steps

9.1 Current Limitations

- Dataset size: 164 clips across 4 subjects is insufficient for robust generalisation. Transformers typically require 100+ subjects for competitive performance.
- Single modality: NIR-only input discards colour information. RGB+NIR fusion would improve robustness under varying illumination.
- No SpO2 / HRV output: the regression head only predicts HR BPM. Clinical utility requires waveform quality sufficient for SpO2 and heart rate variability analysis.
- Motion artefact handling: no explicit MA detection or separation module. ICA / adaptive filtering pre-processing could reduce the NIR channel noise floor.
- Fixed clip length: 160-frame clips assume stationary HR within each window. A sliding-window approach with temporal smoothing would improve BPM tracking for dynamic HR.
- No confidence output: the model predicts a point estimate with no uncertainty quantification. Bayesian or ensemble methods would allow rejection of unreliable clips.

9.2 Recommended Next Steps

1. Expand dataset: Add subjects from UBFC-rPPG, PURE, or DDPM-rPPG to increase training diversity. Target 40+ subjects before clinical benchmarking.
2. Pre-trained initialisation: Initialise the patch embedding from a RGB VideoMAE checkpoint, then fine-tune on NIR. This is expected to reduce the Transformer warm-up from 40 epochs to ~10.
3. Motion Artefact (MA) branch: Add a parallel CNN branch that detects motion-contaminated frames and gates the Transformer attention accordingly (cf. ETA-rPPGNet, CVPR 2023).
4. Frequency-domain loss: Replace L_NegPearson with a spectral loss (FFT peak sharpness around the HR frequency band) — shown to converge faster in Sun et al. (2022).
5. Continuous BPM output: Move from clip-level regression to a sliding-window dense prediction with Kalman filter smoothing for real-time monitoring applications.

10. References

- [1] Yu, Z. et al. (2022). *PhysFormer: Facial Video-Based Physiological Measurement with Temporal Difference Transformer*. IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI). doi:10.1109/TPAMI.2022.3208765 — Core architecture adapted in this work. We modified: channel depth, input channels (3→1), embedding dim (192→96), and added Pre-LN.
- [2] Wang, W. et al. (2017). *Algorithmic principles of remote-PPG*. IEEE Transactions on Biomedical Engineering, 64(7), 1479-1491. — Foundational rPPG theory; bandpass filter design (0.65–4.0 Hz) and peak detection methodology.
- [3] Tsou, Y. et al. (2020). *Multi-Resolution Near-Infrared Face Dataset (MR-NIRP-D)*. IEEE Winter Conf. on Applications of Computer Vision (WACV). — Dataset used in this work. Subject selection, PGM format, and .mat GT format described here.
- [4] Dosovitskiy, A. et al. (2021). *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale*. ICLR 2021. — Tubelet patch embedding design and positional encoding adapted for video.
- [5] Liu, Z. et al. (2022). *Video Swin Transformer*. CVPR 2022. — Spatiotemporal self-attention design patterns referenced for 3D Conv patch extraction.
- [6] Sun, Z. et al. (2022). *Contrast-Phys: Unsupervised Video-Based Remote Physiological Measurement via Spatiotemporal Contrast*. ECCV 2022. — Spectral loss design for rPPG — referenced but not implemented (identified as future work).

End of Submission | PhysFormer-NIR | MR-NIRP-D | GCP / Tesla T4